

CROSS-CUTTING · EXAM STRATEGY - ALL DOMAINS

Federation, App Integration and Managed AI

Customer identity, application front doors, and the services you should not build yourself.

COVERS ALL DOMAINS SCENARIOS 8 MODULE 15 of 22

IAM is for people and services who work **for you**. Cognito is for your **customers**.

Cognito user pool or identity pool?

USER POOL

Authentication. Who are you?

- A directory of your application's users
- Sign-up, sign-in, password reset, MFA
- Social and enterprise identity providers
- Returns JWT tokens the app validates

IDENTITY POOL

Authorisation. What AWS can you touch?

- Exchanges a token for temporary AWS credentials
- Lets an app call S3 or DynamoDB directly
- Can grant limited access to guest users
- Maps identities onto IAM roles

THE TIEBREAKER

'Users must sign in' is a user pool. 'The mobile app must upload directly to S3' is an identity pool. Many designs use both.

Application front doors

SERVICE	USE IT FOR
API Gateway	REST and HTTP APIs, throttling, usage plans, auth, caching
AWS AppSync	Managed GraphQL, with real-time subscriptions and offline sync
Application Load Balancer	HTTP routing to instances, containers or Lambda targets
AWS Amplify	Hosting and back-end scaffolding for web and mobile front ends
CloudFront	Caching and TLS termination in front of any of the above

ON THE EXAM

'Clients request exactly the fields they need' or 'real-time subscriptions' means AppSync, not API Gateway.

Managed AI services, by what they consume

SERVICE	INPUT	OUTPUT
Amazon Rekognition	Images and video	Objects, faces, moderation labels, text in images
Amazon Textract	Scanned documents and forms	Text, tables, and form key-value pairs
Amazon Comprehend	Text	Sentiment, entities, key phrases, PII detection
Amazon Transcribe	Audio	Text transcript, with speaker labels
Amazon Polly	Text	Lifelike speech audio
Amazon Translate	Text	Translated text
Amazon Personalize	User interaction history	Recommendations
Amazon Fraud Detector	Transaction data	Fraud risk scores

ON THE EXAM

Identify these by input and output. 'Extract data from scanned invoices' is Textract, not Rekognition, even though both read images.

A company is launching a consumer mobile application expecting two million users. Users must sign up with an email address, sign in, reset forgotten passwords, and optionally sign in with Google or Apple.

A developer has proposed creating an IAM user for each customer.

What should the solutions architect recommend?

- ☐ A Proceed with IAM users, using groups to manage permissions
- ☐ B Use an Amazon Cognito user pool with social identity providers configured
- ☐ C Use AWS IAM Identity Center federated to a customer directory
- ☐ D Build a custom authentication service backed by DynamoDB

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

- A

 IAM user per customer
IAM is for workforce and workload identity, and its user quota is nowhere near two million. Wrong tool entirely.
- C

 IAM Identity Center
For workforce single sign-on across AWS accounts, not for consumers of your application.
- B

 Cognito user pool
Purpose-built for customer identity at scale, with sign-up, sign-in, password reset, MFA and social providers built in.
- D

 Custom auth service
Builds and owns security-critical authentication code that a managed service already provides.

THE SIGNAL

Customers means Cognito. Employees means IAM Identity Center. Workloads means IAM roles.

A photo-sharing mobile app must let signed-in users upload images directly from their device to S3, without routing the file through an application server.

Each user must be able to write only to their own prefix within the bucket. Users sign in through an existing Cognito user pool.

What should be added?

- ☐ A A Cognito identity pool issuing temporary credentials mapped to an IAM role
- ☐ B Embed IAM access keys in the mobile application
- ☐ C A second Cognito user pool with S3 permissions attached
- ☐ D Make the bucket public and validate uploads afterwards

◆ PAUSE AND DECIDE

What should be added?

- A

 Identity pool with a mapped role
Exchanges the user pool token for temporary AWS credentials, and policy variables can scope each user to their own prefix.
- B

 Embedded access keys
Anyone can extract credentials from a mobile app binary. This is the worst option available.
- C

 A second user pool
User pools authenticate. They do not issue AWS credentials, so no number of them grants S3 access.
- D

 Public bucket
Allows anyone to write anything, and validating afterwards does not prevent the damage.

THE SIGNAL

This is the user pool and identity pool split in one question. Authentication exists already; AWS credentials are what is missing.

An insurance company receives forty thousand scanned claim forms monthly as PDFs. It needs the policy number, claim amount and claimant name extracted into a database. The forms come from several insurers and their layouts differ. The team does not want to train a machine learning model.

Which service should be used?

- ☐ A Amazon Rekognition
- ☐ B Amazon Textract
- ☐ C Amazon Comprehend
- ☐ D Amazon SageMaker with a custom model

◆ PAUSE AND DECIDE

Which service should be used?

- A

Rekognition
Detects objects, faces and text in images, but it does not understand form structure or key-value relationships.
- B

Textract
Built to extract text, tables and form key-value pairs from scanned documents, including layouts it has not seen before.
- C

Comprehend
Analyses text that already exists. It cannot read a scanned image to produce that text.
- D

SageMaker custom model
Requires training, labelling and ongoing maintenance, which the question explicitly rules out.

THE SIGNAL

Rekognition sees images. Textract reads documents. Comprehend understands text it is given. Structured forms are always Textract.

A mobile application must show a live scoreboard that updates on every connected device within a second of a change. Clients also request different subsets of fields depending on the screen they are on, and must work offline with synchronisation on reconnect.

The team wants a managed service rather than running WebSocket servers.

Which service should be used?

- ☐ A AWS AppSync
- ☐ B Amazon API Gateway REST APIs with client polling
- ☐ C An Application Load Balancer with EC2 WebSocket servers
- ☐ D Amazon SNS with mobile push notifications

◆ PAUSE AND DECIDE

Which service should be used?

- A

AppSync

Managed GraphQL with real-time subscriptions, field-level queries and offline sync. All three requirements in one service.
- B

REST with polling

Polling every second from every device is wasteful, and REST returns fixed shapes rather than requested field subsets.
- C

Self-managed WebSocket servers

Exactly the infrastructure the team said it does not want to run.
- D

SNS push notifications

Delivers notifications to devices. It is not a data API and offers no field selection or offline sync.

THE SIGNAL

Real-time subscriptions plus clients choosing their own fields plus offline sync is the AppSync signature.

A contact centre records one hundred thousand customer calls monthly. Compliance needs searchable transcripts, and the customer experience team wants to know which calls contained frustration or complaints.

The company does not want to build or train models.

Which combination meets these requirements? (Choose two.)

- ☐ A Amazon Transcribe to convert the recordings to text
- ☐ B Amazon Comprehend for sentiment analysis on the transcripts
- ☐ C Amazon Polly to process the audio
- ☐ D Amazon Rekognition to analyse the call recordings

◆ PAUSE AND DECIDE

Which combination meets these requirements? (Choose two.)

- | | |
|--|---|
| <div data-bbox="173 439 234 492">A</div> <div data-bbox="257 446 473 485">Transcribe</div> <div data-bbox="257 508 1059 591">Converts recorded audio into text, producing the searchable transcripts compliance requires.</div> | <div data-bbox="1304 439 1365 492">B</div> <div data-bbox="1388 446 1663 485">Comprehend</div> <div data-bbox="1388 508 2333 591">Runs sentiment analysis over the resulting text, identifying frustration and complaints.</div> |
| <div data-bbox="173 629 234 682">C</div> <div data-bbox="257 636 351 675">Polly</div> <div data-bbox="257 698 1156 779"><i>Turns text into speech. That is the opposite direction to what is needed here.</i></div> | <div data-bbox="1304 629 1365 682">D</div> <div data-bbox="1388 636 1625 675">Rekognition</div> <div data-bbox="1388 698 2318 736"><i>Analyses images and video. These are audio recordings.</i></div> |

THE SIGNAL

A two-stage pipeline: audio to text with Transcribe, then text to meaning with Comprehend. Polly runs the other way.

A company's employees must sign in to a new internal web application using their existing corporate Microsoft Entra ID credentials. The application runs on EC2 behind an Application Load Balancer.

The team does not want to manage a separate set of application passwords.

Which approach meets this requirement?

- ☐ A A Cognito user pool federated to the corporate identity provider, with ALB authentication
- ☐ B Create IAM users for every employee and issue access keys
- ☐ C A Cognito identity pool with unauthenticated access enabled
- ☐ D Store employee passwords in AWS Secrets Manager

◆ PAUSE AND DECIDE

Which approach meets this requirement?

- A** Federated user pool with ALB auth
The ALB can authenticate users against a Cognito user pool, which federates to the corporate provider, so no new passwords exist.
- B** IAM users with access keys
IAM controls access to AWS APIs, not sign-in to your own web application.
- C** Identity pool, unauthenticated
Unauthenticated access means no sign-in at all, which is the opposite of the requirement.
- D** Passwords in Secrets Manager
Creates the separate password set the team wants to avoid, and Secrets Manager is not an authentication service.

THE SIGNAL

Application load balancers can authenticate through Cognito or OIDC before traffic ever reaches your instances.

A retailer wants product recommendations on its website based on each customer's browsing and purchase history. It has three years of interaction data.

The team has no machine learning specialists and wants results within a few weeks.

Which service should be used?

- ☐ A Amazon Personalize
- ☐ B Amazon SageMaker with a custom recommendation model
- ☐ C Amazon Comprehend on product descriptions
- ☐ D Amazon Forecast on sales volumes

◆ PAUSE AND DECIDE

Which service should be used?

- A

 Amazon Personalize
Built specifically to produce recommendations from interaction history, with no ML expertise required.
- B

 SageMaker custom model
The most capable option and the one needing specialists the team does not have, on a timeline that does not allow it.
- C

 Comprehend
Analyses text for sentiment and entities. It does not generate recommendations.
- D

 Amazon Forecast
Predicts future values of a time series, such as demand. That is forecasting, not personalised recommendation.

THE SIGNAL

Personalize recommends to a user. Forecast predicts a quantity over time. Both use history and answer different questions.

A social platform must automatically detect and block uploaded images containing nudity or graphic violence before they appear publicly. Volume is around two hundred thousand images daily.

The company wants a managed service and needs a confidence score so borderline cases can go to human review.

Which service should be used?

- | | |
|--|--|
| ☐ A Amazon Rekognition content moderation | ☐ B Amazon Textract |
| ☐ C Amazon Macie | ☐ D Amazon Comprehend |

◆ PAUSE AND DECIDE

Which service should be used?

- A

 Rekognition content moderation
Returns moderation labels with confidence scores for images and video, which supports an automatic block plus human review threshold.
- B

 Textract
Extracts text and form data from documents. It does not classify image content.
- C

 Macie
Finds sensitive data such as personal identifiers inside S3 objects. It does not assess images for graphic content.
- D

 Comprehend
Analyses text. These are images.

THE SIGNAL

Confidence scores plus a human review threshold is a recurring pattern. The service supplies the score; you choose the cut-off.

Module 15 takeaways

- ◆ **Customers mean Cognito.** Employees mean IAM Identity Center. Workloads mean IAM roles.
- ◆ **User pool authenticates. Identity pool issues AWS credentials.** Many designs need both.
- ◆ **Rekognition sees images. Textract reads documents.** Structured forms are always Textract.
- ◆ **Transcribe is audio to text. Polly is text to audio.** Opposite directions.
- ◆ **Real-time subscriptions plus field selection plus offline sync** means AppSync.

ON THE EXAM

Next: the confusion pairs module, which gathers the sharpest distinctions from the whole course into one place.